



Lab Number & Title: Lab 01: Ubuntu Server Installation and Environment Setup

Student Name: Umaiza Waseem

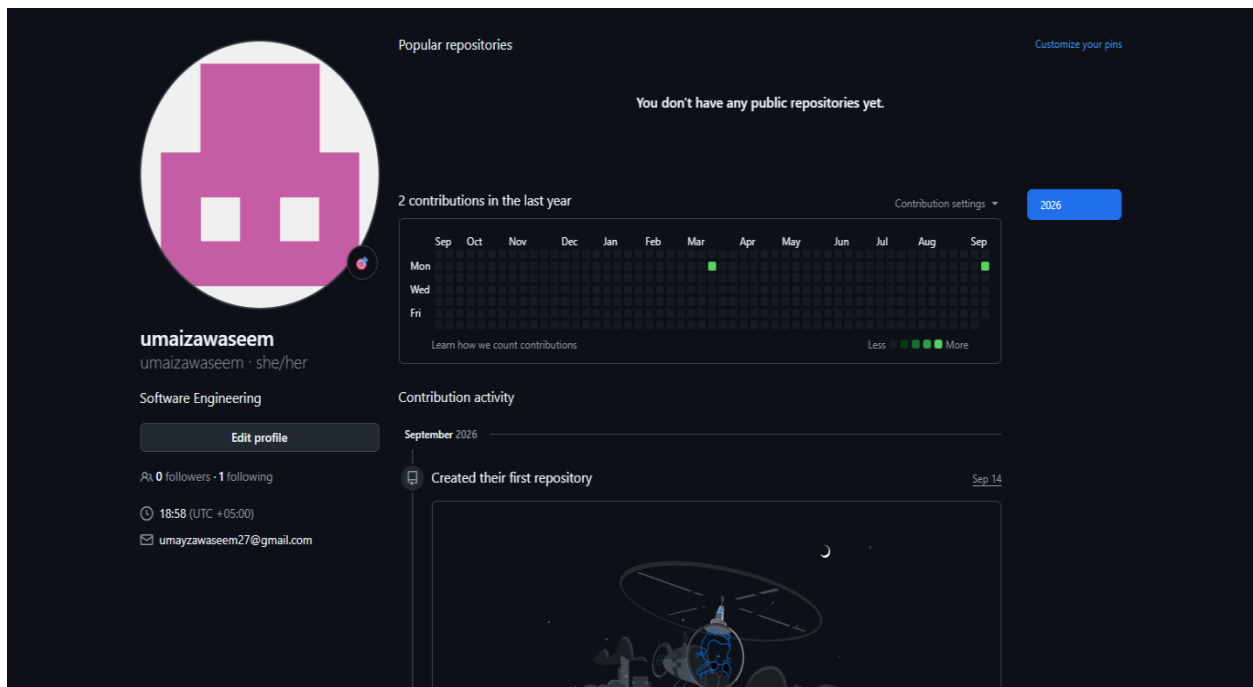
Registration Number: FA24-B1-SE-083

Course Code: SE-665— Cloud Computing

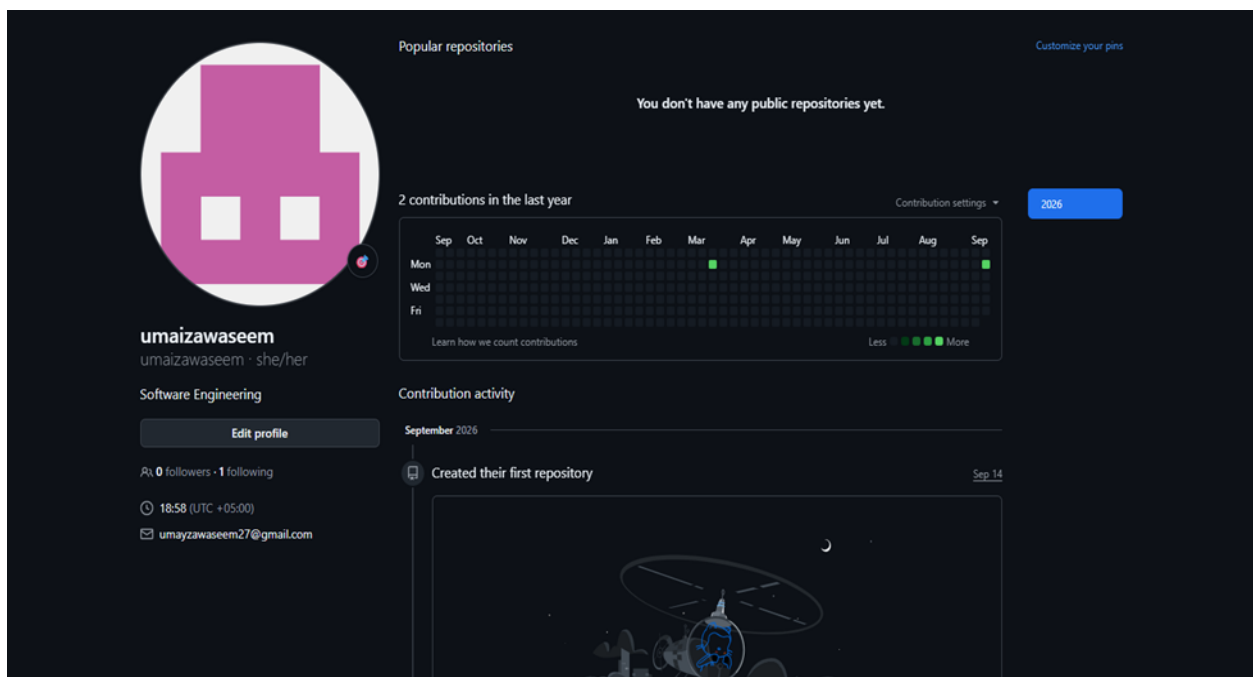
Section: Section A

GitHub Username: Umaiza

Task 1: Create and Configure Your GitHub Account

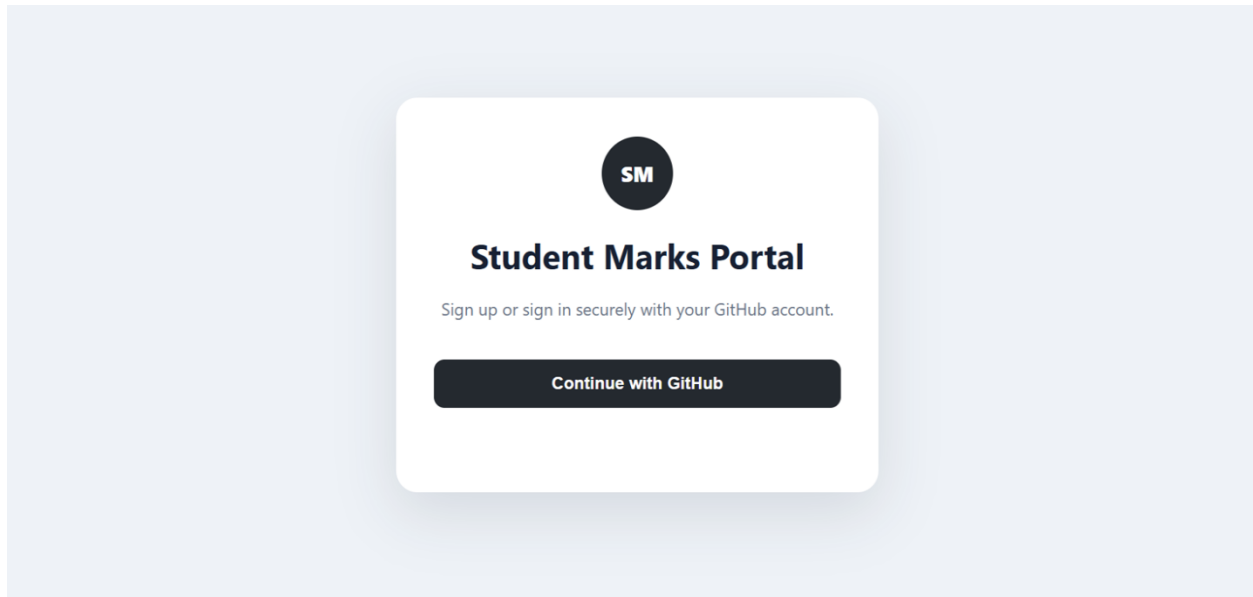


Profile overview showing the newly registered GitHub account

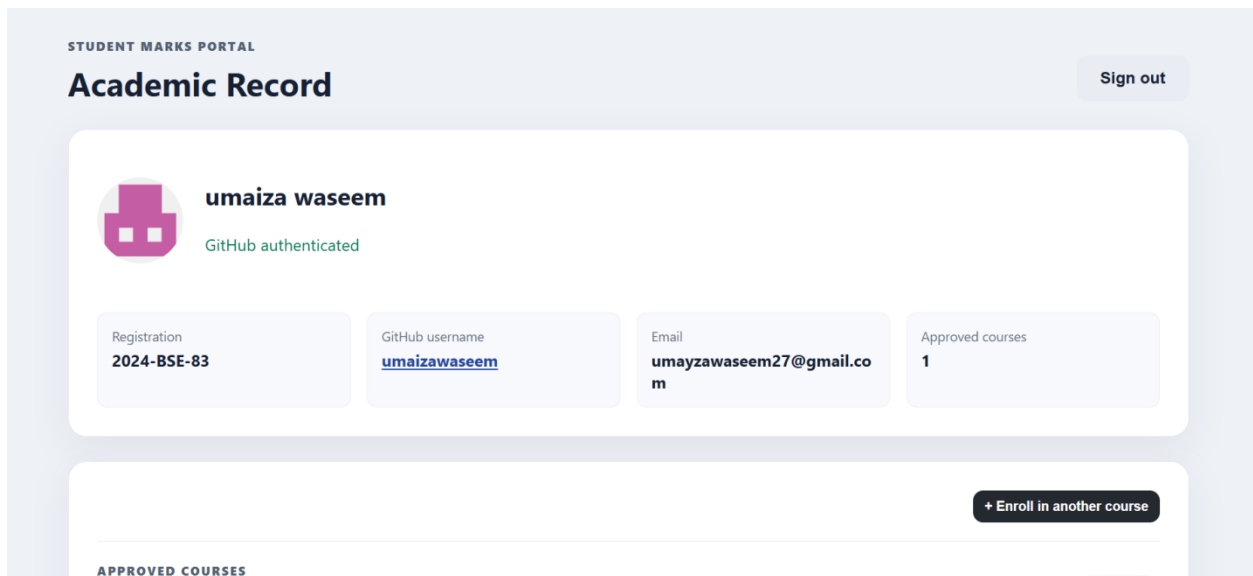


Shows the updated GitHub profile details including display name and contact email

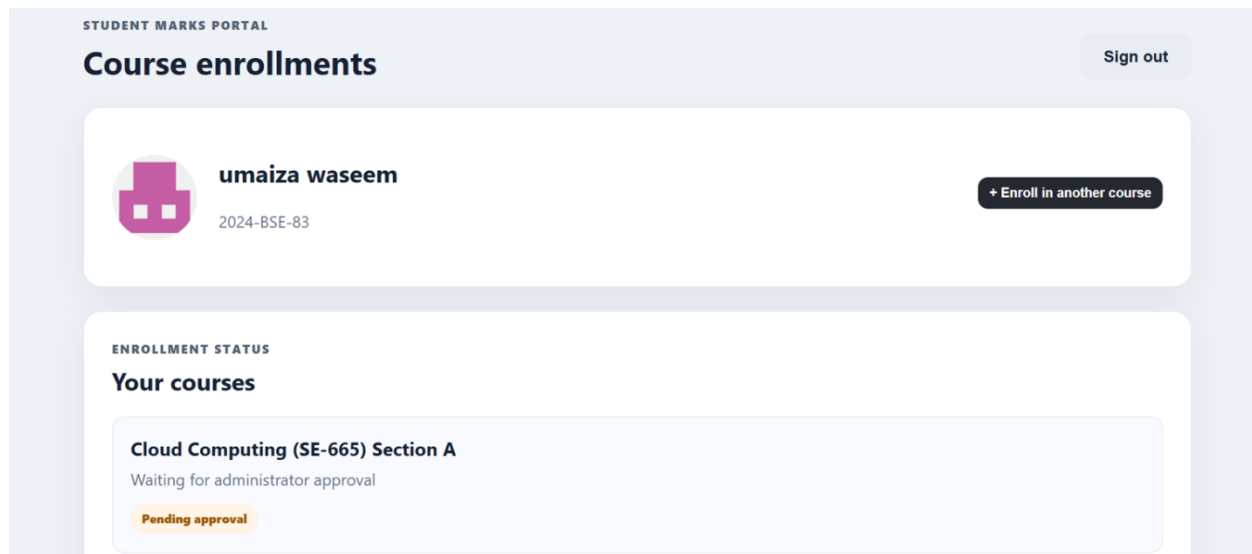
Task 2: Register in the Student Marks Portal



Sign-in landing page for the Student Marks Portal using GitHub OAuth.

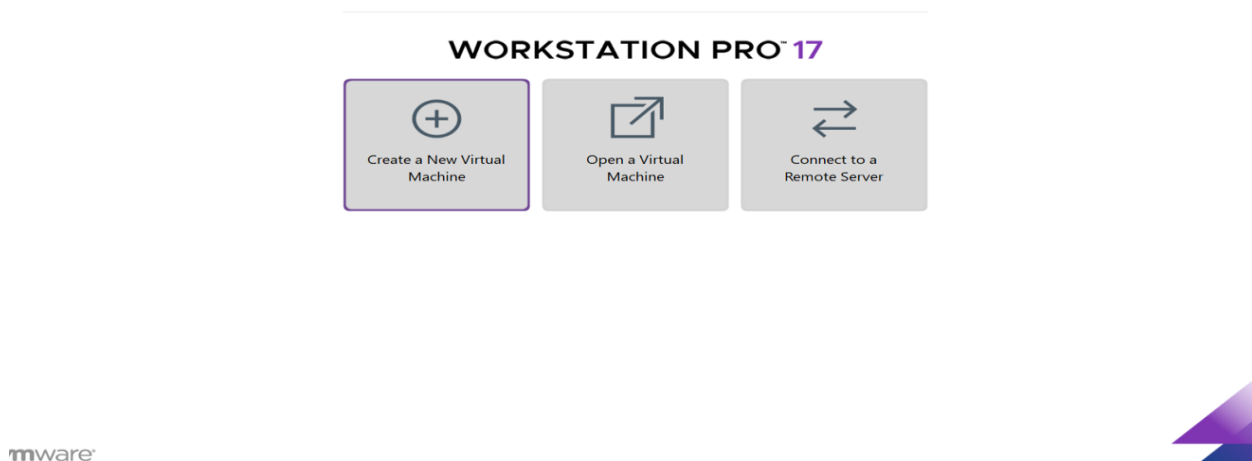


Registration form filled with registration number 2024-BSE-64, course SE-665 — Cloud Computing, and Section A



Shows pending course enrollment approval for Cloud Computing Section A.

Task 3: Download VMware Workstation Pro and Ubuntu Server



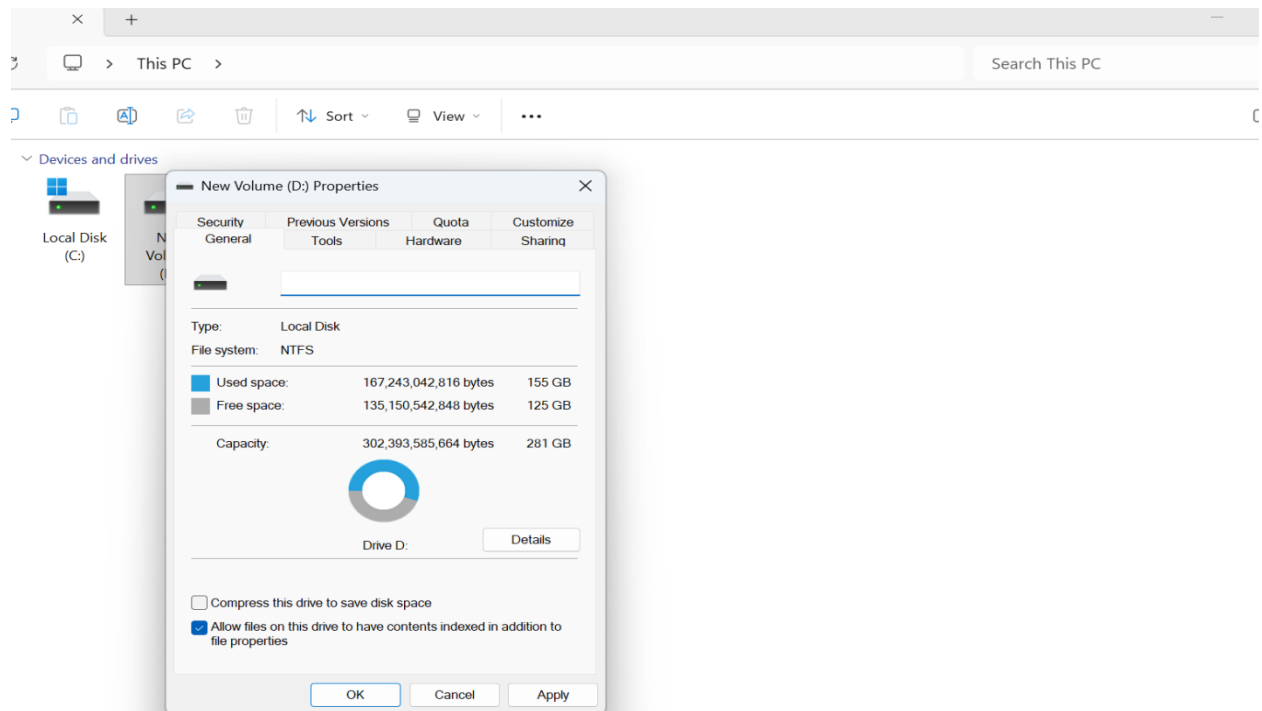
VMware Workstation Pro home screen

New folder (2)	9/13/2026 1:39 AM	File folder	
ubuntu-22.04.5-live-server-amd64	9/13/2026 6:23 PM	Disc Image File	2,086,842 KB
Unconfirmed 477125.crdownload	9/13/2026 6:48 PM	CRDOWNLOAD File	1,200 KB
ubuntu-26.04.1-live-server-amd64	9/13/2026 7:51 PM	Disc Image File	2,859,240 KB

Yesterday

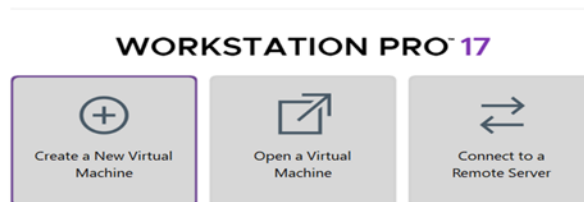
Type: Disc Image File

Downloaded Ubuntu 26.04 live server ISO file in local downloads directory



Verification of available disk storage for virtual machine installation.

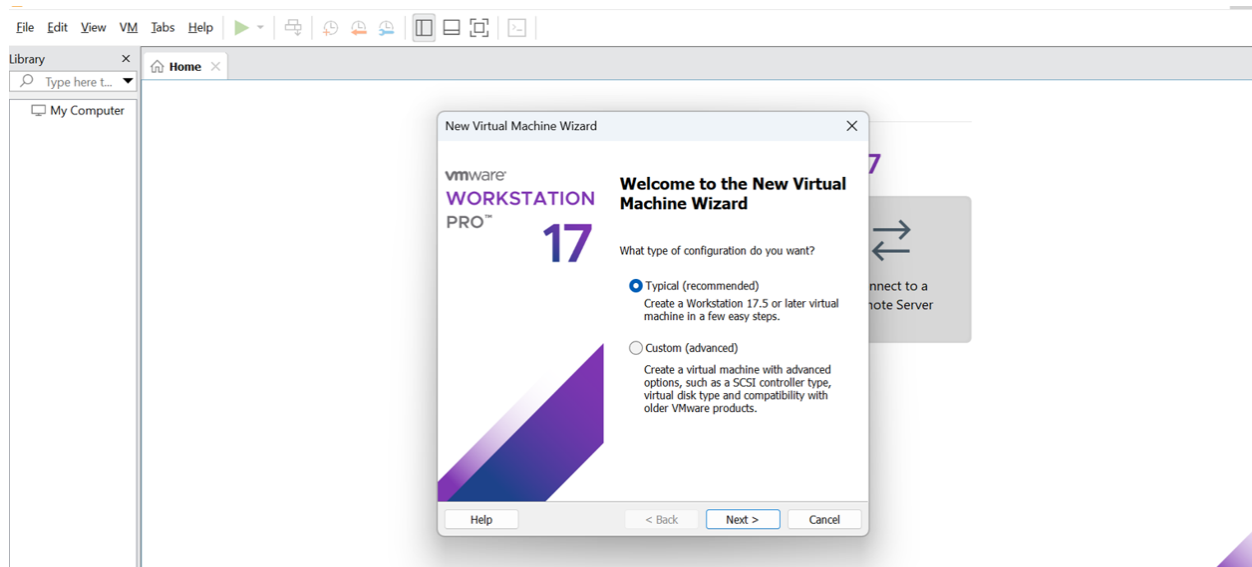
Task 4: Create the Ubuntu Server Virtual Machine



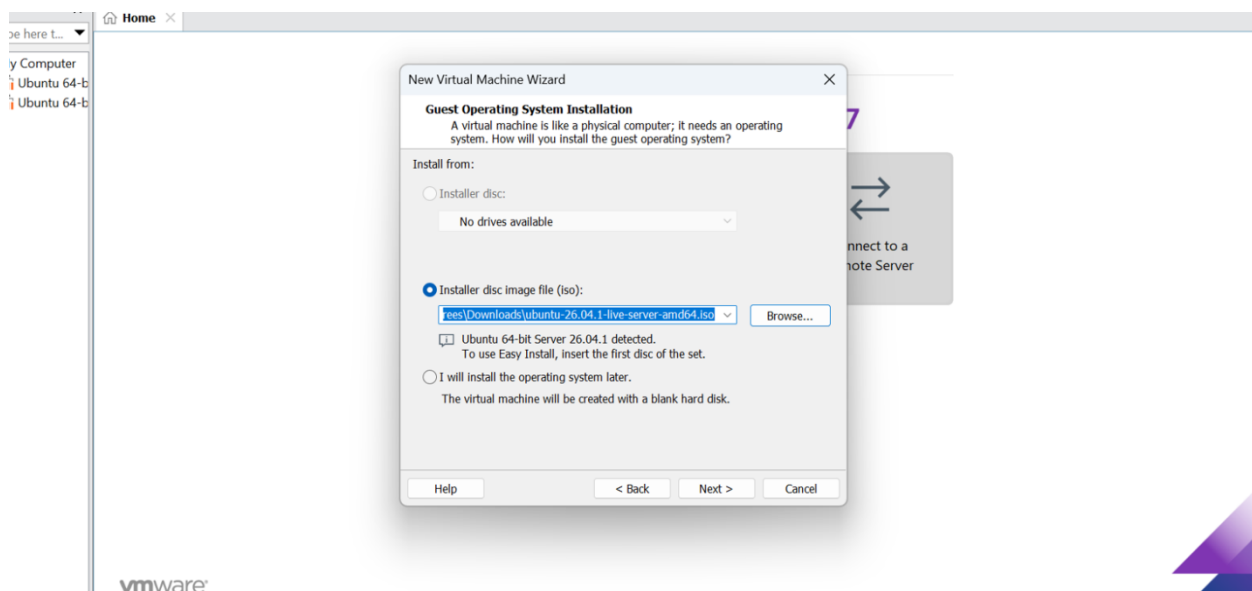
vmware



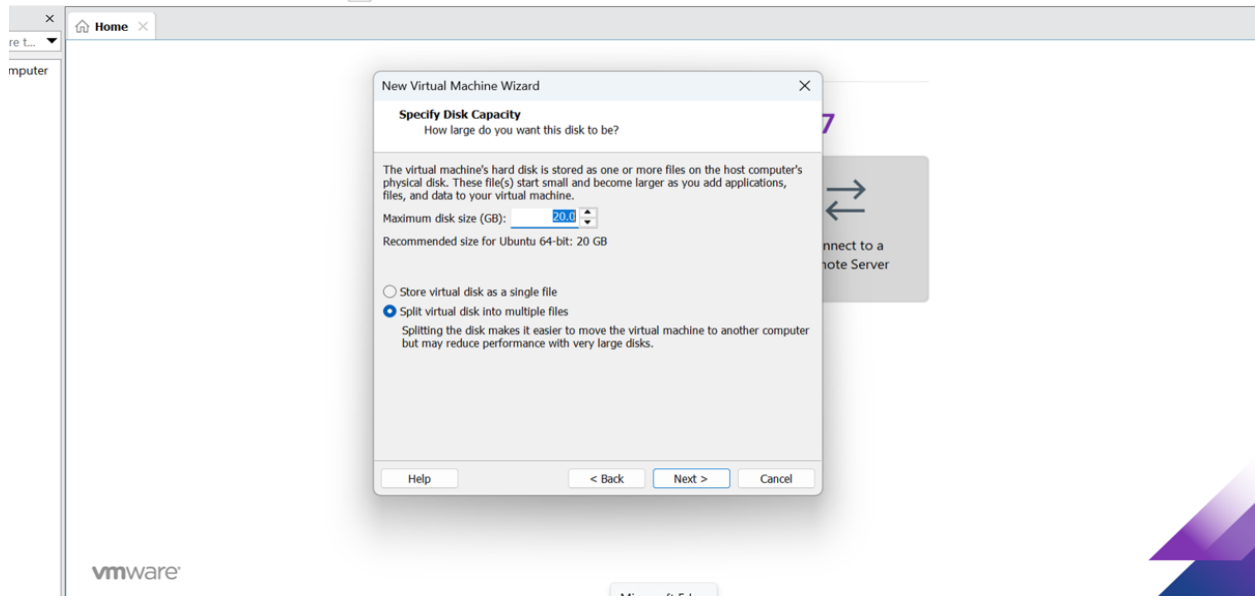
Selecting "Create a New Virtual Machine" in VMware Workstation



Selecting Typical configuration mode for virtual machine creation.



Browsing and selecting the target Ubuntu Server ISO image

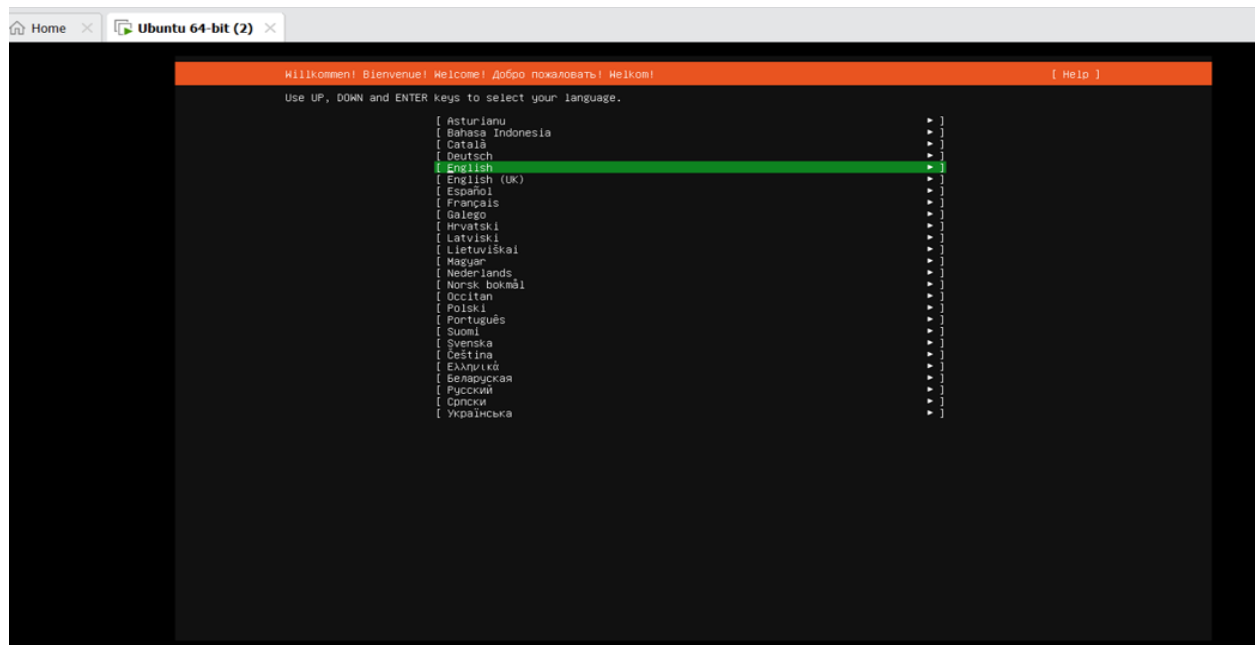


Setting virtual disk size to 20.0 GB and selecting split disk storage



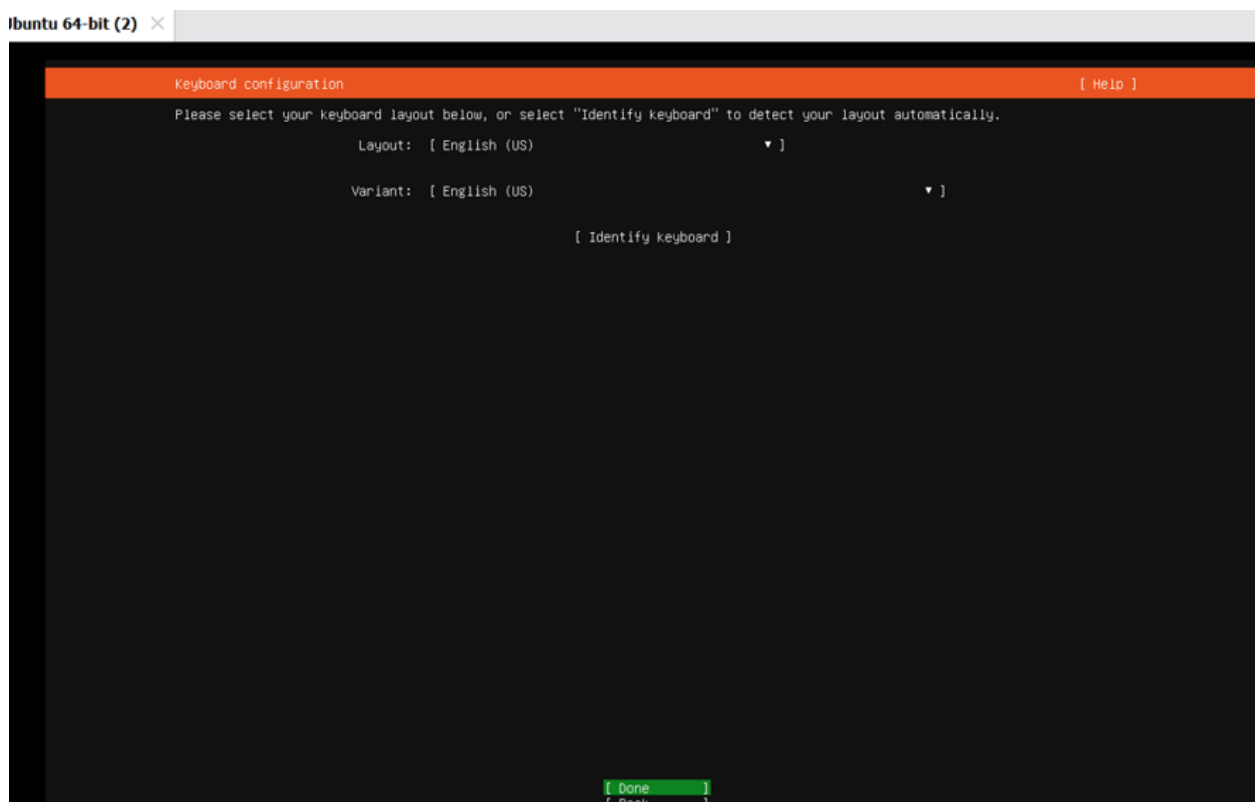
First boot of the virtual machine entering the Ubuntu installer GRUB menu.

Task 5: Choose the Installation Language

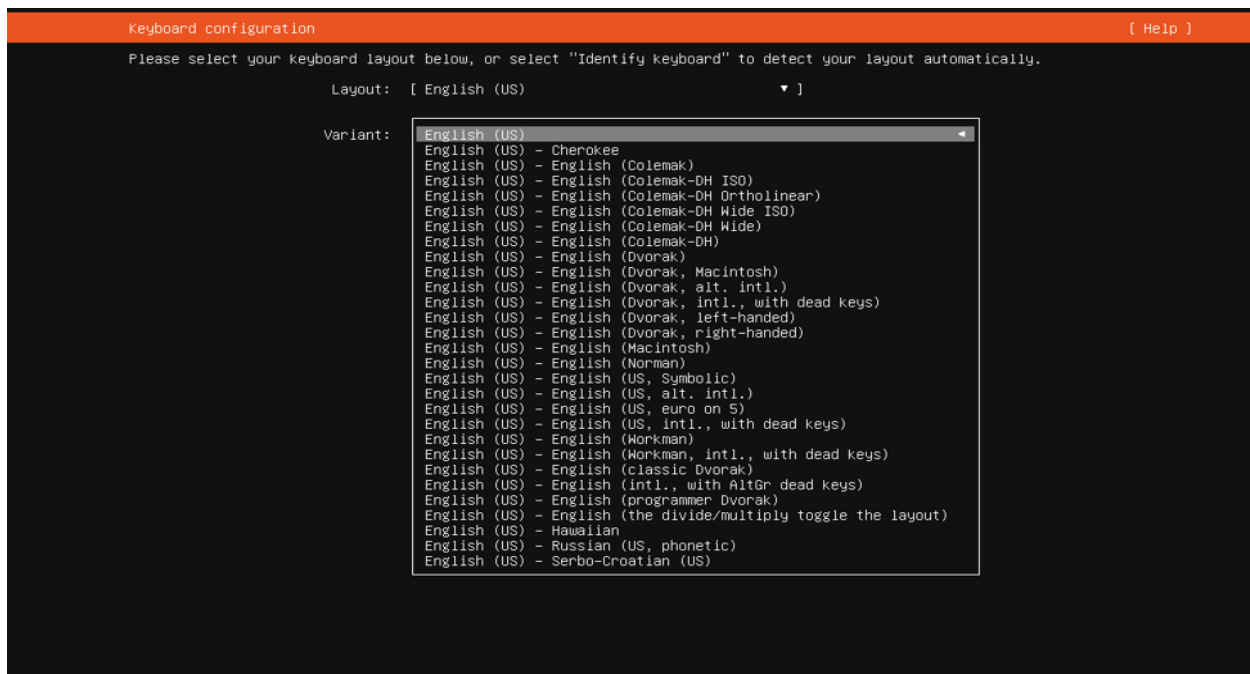


Language selection screen in the Ubuntu installer.

Task 6: Configure the Keyboard Layout

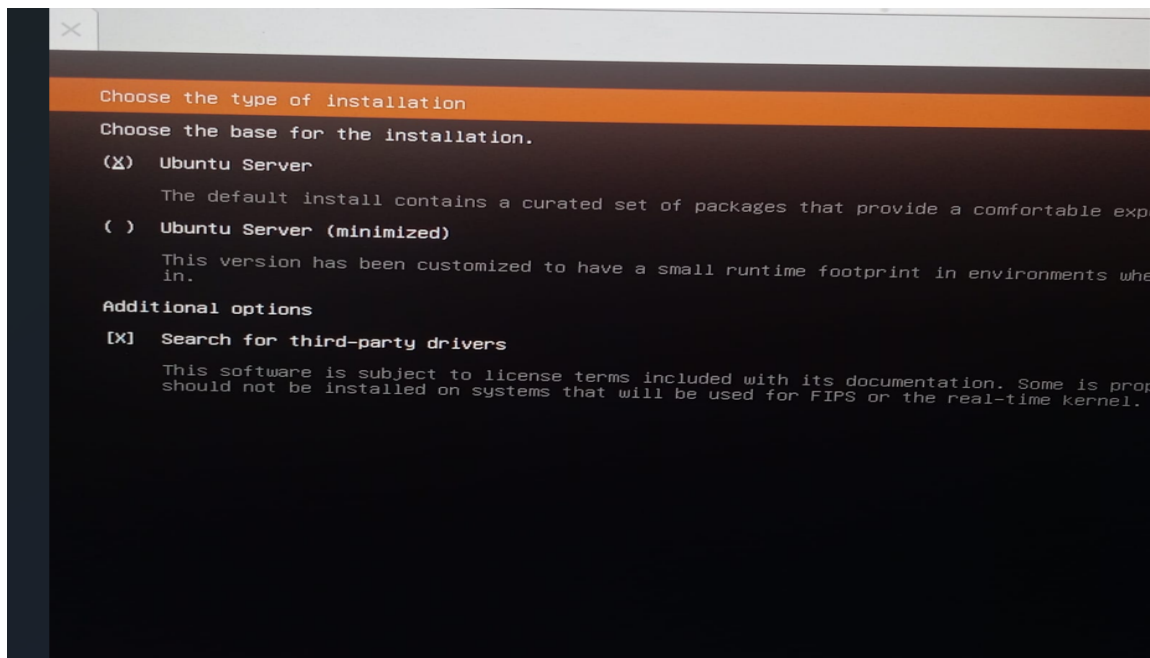


Choosing primary keyboard layout as English (US)



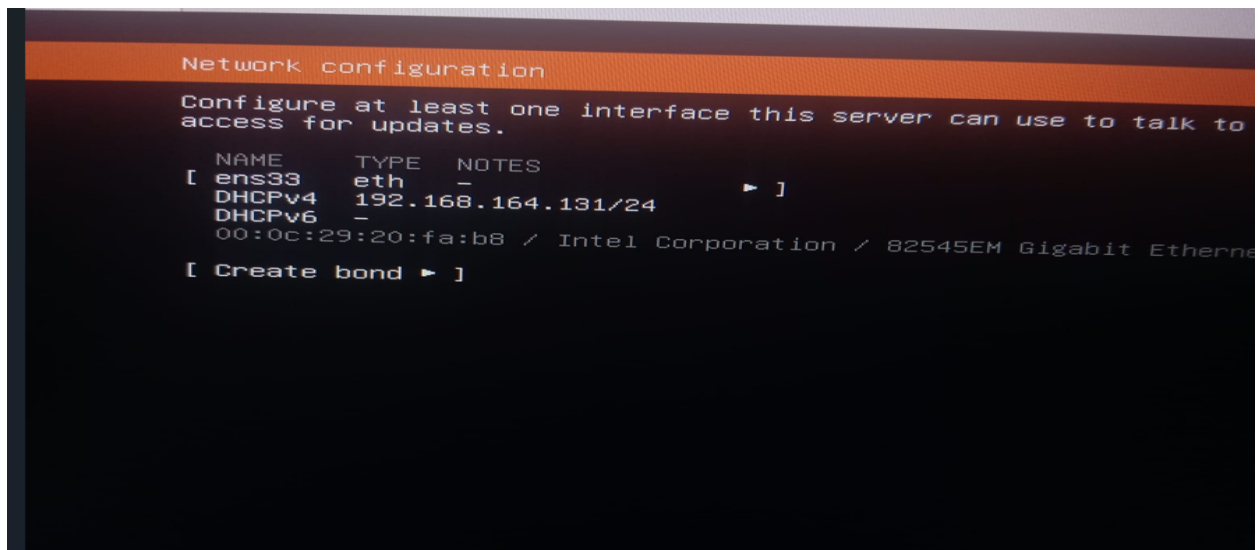
Confirming keyboard variant selection as English (US).

Task 7: Choose the Ubuntu Server Installation Type

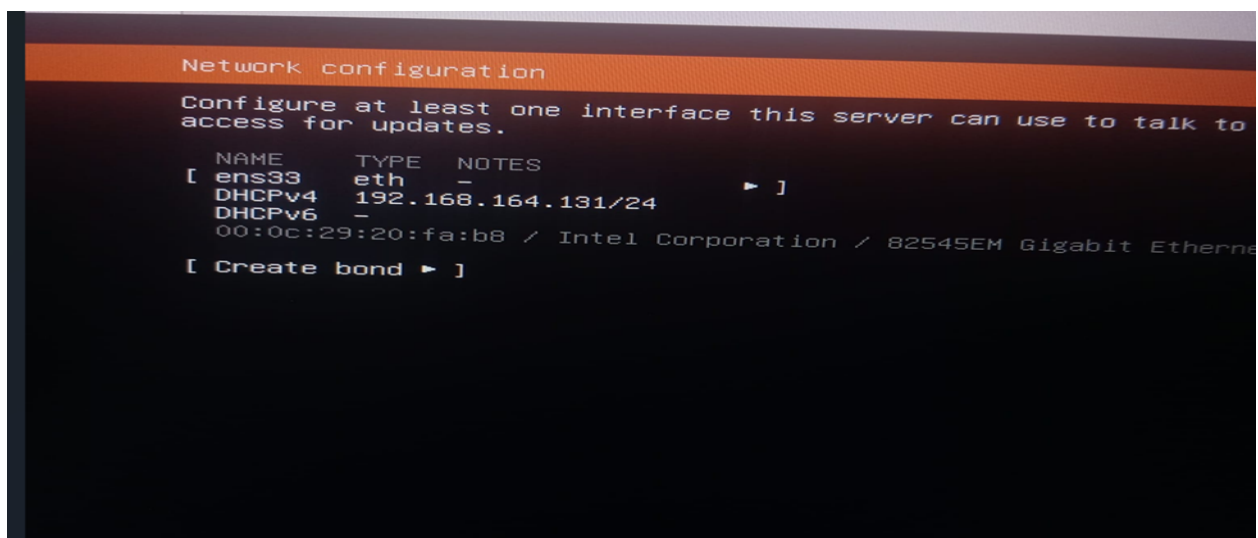


Selecting standard Ubuntu Server installation option.

Task 8: Configure the Network

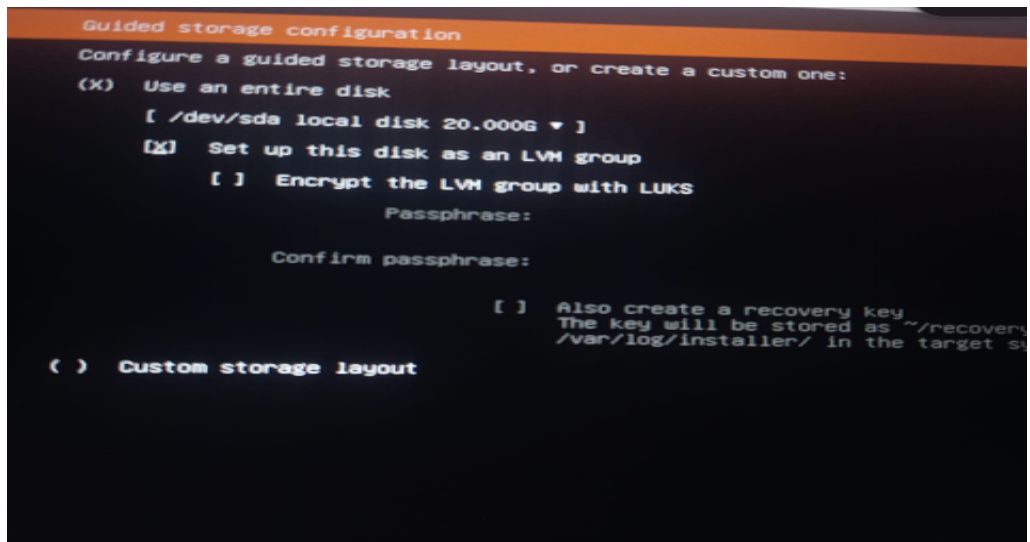


Automatic DHCP network IP assignment on interface ens33

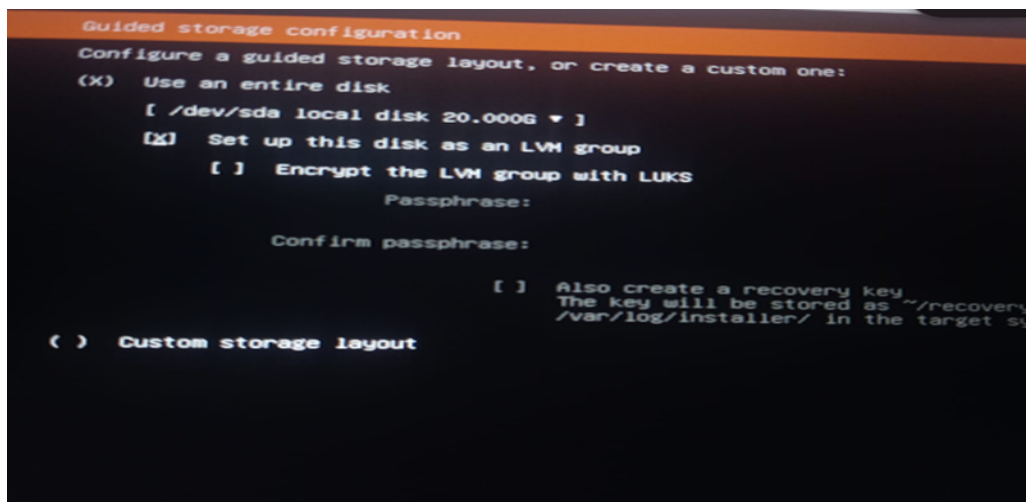


Active Ethernet connection confirmation

Task 9: Configure Storage

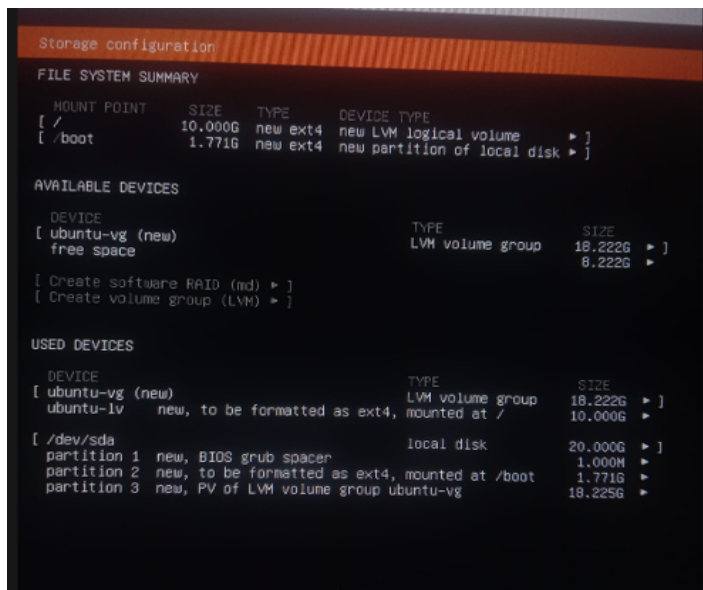


Configuring storage layout using entire 20GB disk with LVM group enabled.

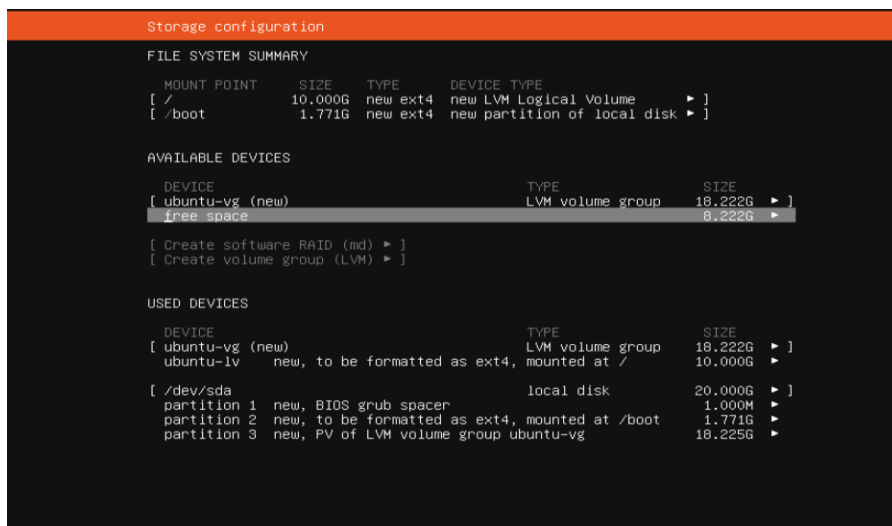


Finalized guided storage selection before filesystem generation

Task 10: Select the Installation Disk

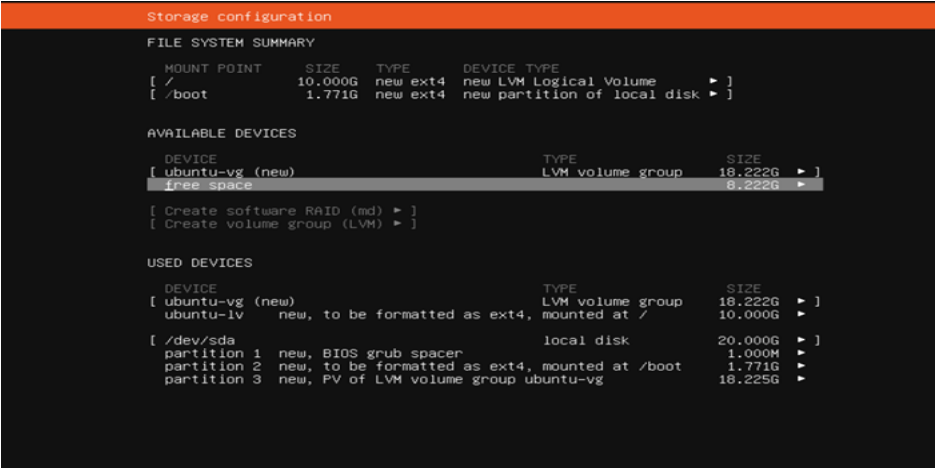


Summary view of mount points, root partition (/), and /boot configuration



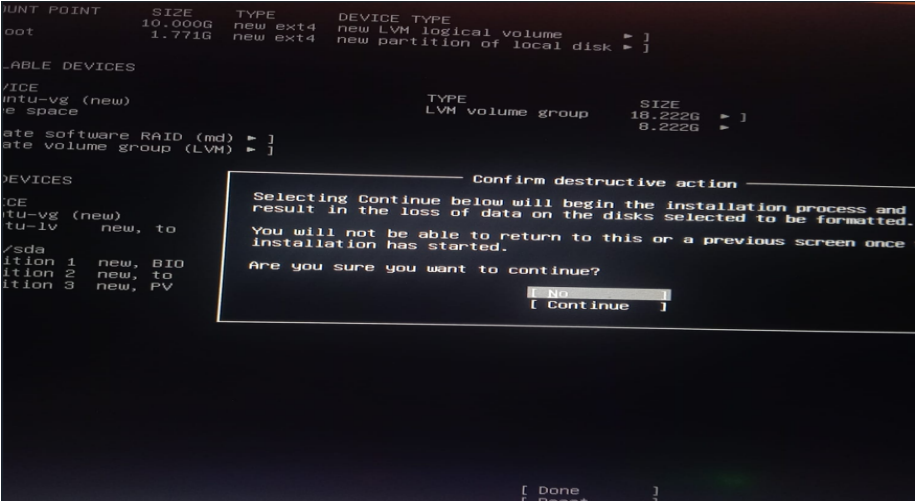
Disk partition allocation map showing LVM volume group storage breakdown.

Task 11: Review the Partitions

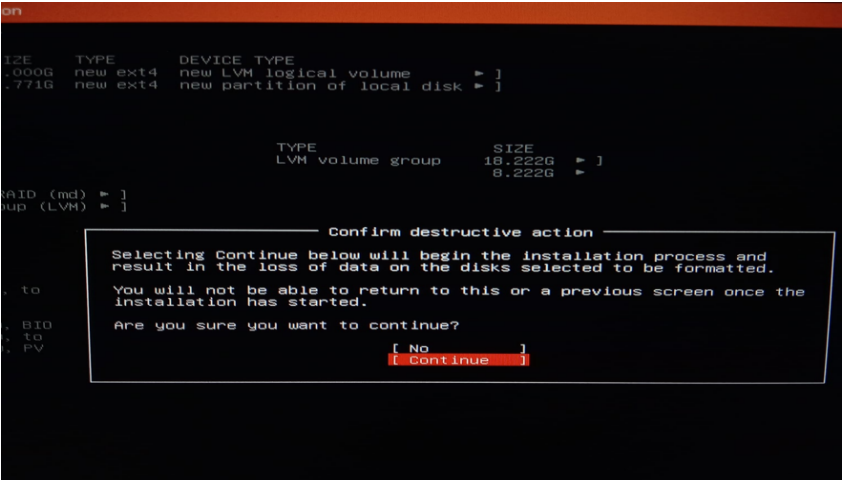


Final review of device mappings, LVM logical volumes, and ext4 formatting.

Task 12: Confirm the Storage Changes

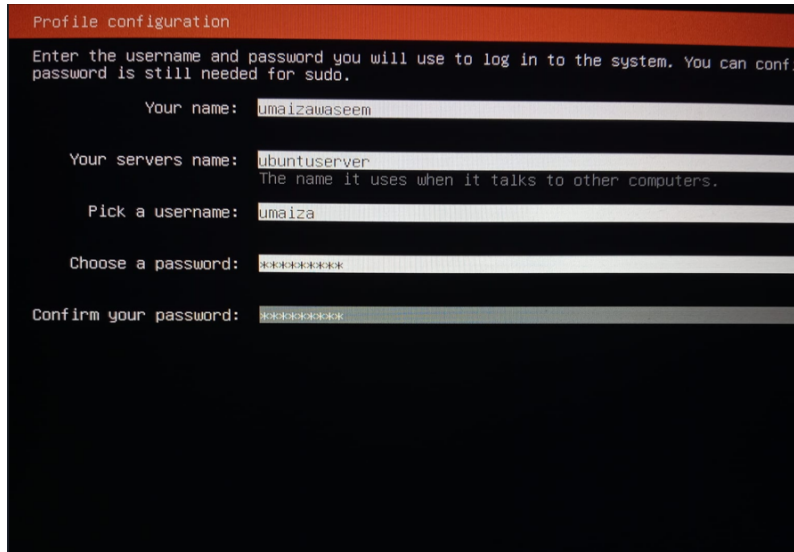


Confirmation pop-up warning about disk formatting.



Selecting Continue to execute disk partitioning and formatting

Task 13: Configure the Ubuntu Profile



The screenshot shows the 'Profile configuration' window with a red header. Below the header, instructions state: 'Enter the username and password you will use to log in to the system. You can configure the system after the installation. A root password is still needed for sudo.' The form contains five input fields with the following values: 'Your name:' is 'umaizawaseem'; 'Your servers name:' is 'ubuntu-server' with a subtext 'The name it uses when it talks to other computers.'; 'Pick a username:' is 'umaiza'; 'Choose a password:' is masked with '*****'; and 'Confirm your password:' is also masked with '*****'.

```
Profile configuration
Enter the username and password you will use to log in to the system. You can configure the system after the installation. A root password is still needed for sudo.

Your name: umaizawaseem

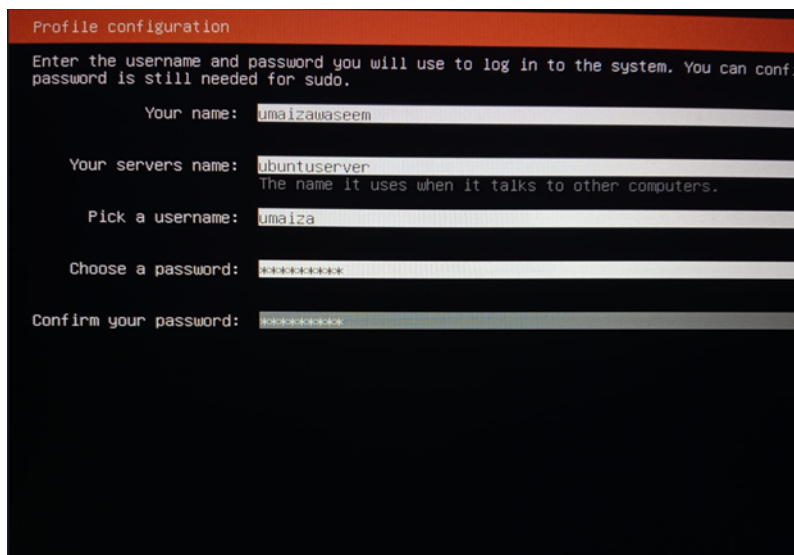
Your servers name: ubuntu-server
The name it uses when it talks to other computers.

Pick a username: umaiza

Choose a password: *****

Confirm your password: *****
```

Real name field populated.



This screenshot is identical to the one above, showing the 'Profile configuration' window with the same instructions and populated fields: 'Your name: umaizawaseem', 'Your servers name: ubuntu-server', 'Pick a username: umaiza', 'Choose a password: *****', and 'Confirm your password: *****'.

```
Profile configuration
Enter the username and password you will use to log in to the system. You can configure the system after the installation. A root password is still needed for sudo.

Your name: umaizawaseem

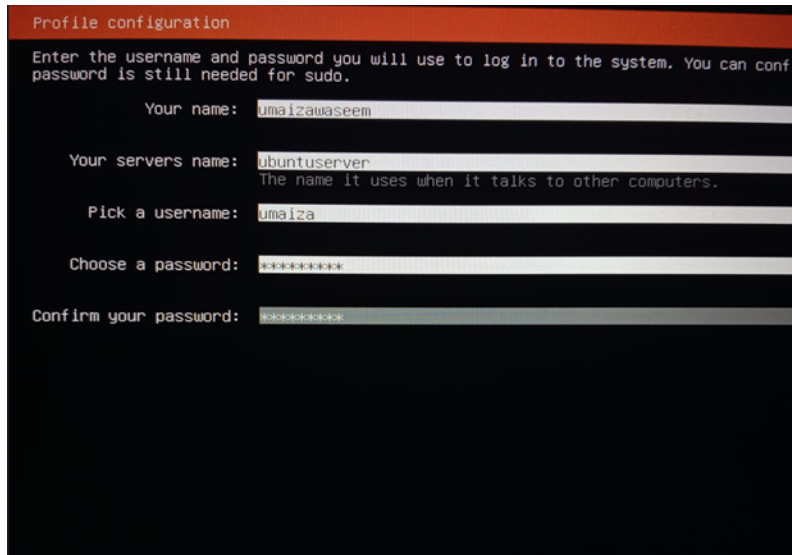
Your servers name: ubuntu-server
The name it uses when it talks to other computers.

Pick a username: umaiza

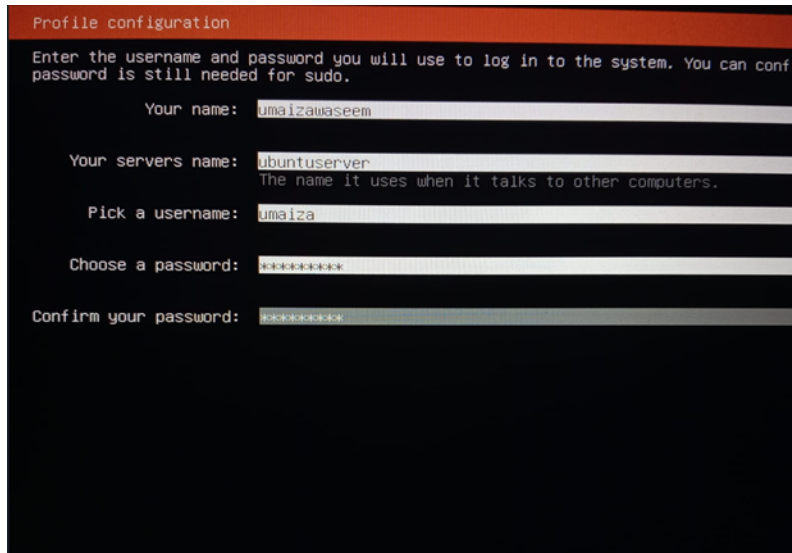
Choose a password: *****

Confirm your password: *****
```

Setting server hostname to ubuntu



Configuring user account login name as nibasadiq.



Entering and confirming user account password.

Task 14: Complete the Software Installation


```
Installing system

subiquity/2dev/apply_autoinstall_config:
subiquity/Ad/apply_autoinstall_config:
subiquity/Late/apply_autoinstall_config:
configuring apt
  curtin command in-target
installing system
  executing curtin install initial step
  executing curtin install partitioning step
    curtin command install
      configuring storage
        running 'curtin block-meta simple'
        curtin command block-meta
          removing previous storage devices
          configuring disk: disk-sda
          configuring partition: partition-0
          configuring partition: partition-1
          configuring partition: partition-2
          configuring format: format-0
          configuring lvm_voigroup: lvm_voigroup-0
          configuring lvm_partition: lvm_partition-0
          configuring format: format-1
          configuring mount: mount-1
          configuring mount: mount-0
      executing curtin install extract step
        curtin command install
          writing install sources to disk
          running 'curtin extract'
          curtin command extract
            acquiring and extracting image from cp:///tmp/tmp5ria4dy/mount
      configuring keyboard
      curtin command in-target
    executing curtin install curthooks step
      curtin command install
        configuring installed system
        running 'curtin curthooks'
        curtin command curthooks
        configuring apt configuring apt
        installing missing packages
```

Base system installation log showing automated package deployment and storage setup

Task 15: Reboot and Verify the Ubuntu Identity

```
Finalizing installation
  running 'curtin hook'
  curtin command hook
  executing late commands

[ View full log ]
[ Reboot Now ]
```

Selecting "Reboot Now" after installation finishes


```
Command Prompt
Microsoft Windows [Version 10.0.26200.8655]
(c) Microsoft Corporation. All rights reserved.

C:\Users\waseem.idrees>
C:\Users\waseem.idrees>
C:\Users\waseem.idrees> ssh-keygen -t ed25519 -C "umayzawaseem27@gmail.com"
Generating public/private ed25519 key pair.
Enter file in which to save the key (C:\Users\waseem.idrees/.ssh/id_ed25519):
Created directory 'C:\Users\waseem.idrees/.ssh'.
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in C:\Users\waseem.idrees/.ssh/id_ed25519
Your public key has been saved in C:\Users\waseem.idrees/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:pz7DjGfPK7gRVsAq3aYbWbnlxAfPsdtdYZvo3DggI6A umayzawaseem27@gmail.com
The key's randomart image is:
+--[ED25519 256]--+
|
| . . .
| . o o + + = +
| E . + X = * +
| . B XS*.*
| + * =o* +
| + X.. .
| . o.*
| .o
+-----[SHA256]-----+
C:\Users\waseem.idrees>
```

```
Command Prompt
Enter same passphrase again:
Your identification has been saved in C:\Users\waseem.idrees/.ssh/id_ed25519
Your public key has been saved in C:\Users\waseem.idrees/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:pz7DjGfPK7gRVsAq3aYbWbnlxAfPsdtdYZvo3DggI6A umayzawaseem27@gmail.com
The key's randomart image is:
+--[ED25519 256]--+
|
| . . .
| . o o + + = +
| E . + X = * +
| . B XS*.*
| + * =o* +
| + X.. .
| . o.*
| .o
+-----[SHA256]-----+

C:\Users\waseem.idrees>type C:\Users\waseem.idrees\.ssh\id_ed25519.pubtype C:\Users\waseem.idrees\.ssh\id_ed25519.pub
The system cannot find the file specified.
Error occurred while processing: C:\Users\waseem.idrees\.ssh\id_ed25519.pubtype.

C:\Users\waseem.idrees\.ssh\id_ed25519.pub

ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIBStVp4+Jaw2aQQHcTLXTMQLkviM8VG9gqW4VFQpWwMv umayzawaseem27@gmail.com
C:\Users\waseem.idrees>
C:\Users\waseem.idrees> type C:\Users\waseem.idrees\.ssh\id_ed25519.pub
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIBStVp4+Jaw2aQQHcTLXTMQLkviM8VG9gqW4VFQpWwMv umayzawaseem27@gmail.com
C:\Users\waseem.idrees>
C:\Users\waseem.idrees>
```

System cloud-init process generating SSH host keys on first boot.

Task 16: Find the Ubuntu Server IP Address

```

applicable law.
umaiza@ubuntu:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:20:fa:b8 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altnamename enx00c2920fab8
    inet 192.168.164.132/24 metric 100 brd 192.168.164.255 scope global dynamic ens33
        valid_lft 1691sec preferred_lft 1691sec
    inet6 fe80::20c:29ff:fe20:fab8/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
umaiza@ubuntu:~$
umaiza@ubuntu:~$
umaiza@ubuntu:~$
umaiza@ubuntu:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:20:fa:b8 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altnamename enx00c2920fab8
    inet 192.168.164.132/24 metric 100 brd 192.168.164.255 scope global dynamic ens33
        valid_lft 1679sec preferred_lft 1679sec
    inet6 fe80::20c:29ff:fe20:fab8/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
umaiza@ubuntu:~$ ssh umaiza@192.
ssh: Could not resolve hostname 192.: Temporary failure in name resolution
umaiza@ubuntu:~$
umaiza@ubuntu:~$
umaiza@ubuntu:~$
umaiza@ubuntu:~$ ssh umaiza@192.168.164.132
The authenticity of host '192.168.164.132 (192.168.164.132)' can't be established.
ED25519 key fingerprint is: SHA256:L8vbU16VbbZHJ7GVdGcg5FUIEkzLgkBXqvs39dTJ4XA
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.164.132' (ED25519) to the list of known hosts.
umaiza@192.168.164.132's password:
Connection closed by 192.168.164.132 port 22
umaiza@ubuntu:~$

```

Output of ip addr showing network interfaces and assigned IPv4 address 192.168.247.133 on ens33.

Task 17: Connect to Ubuntu Server from Windows

```

umaiza@ubuntu: ~
Microsoft Windows [Version 10.0.26200.8655]
(c) Microsoft Corporation. All rights reserved.

C:\Users\waseem.idrees>ssh umaiza@192.168.164.132
The authenticity of host '192.168.164.132 (192.168.164.132)' can't be established.
ED25519 key fingerprint is SHA256:L8vbU16VbbZHJ7GVdGcg5FUIEkzLgkBXqvs39dTJ4XA.
This key is not known by any other names.

```

Windows command prompt executing SSH command

```

Are you sure you want to continue connecting (yes/no/[fingerprint])?
Host key verification failed.
C:\Users\waseem.idrees>ssh umaiza@192.168.164.132
The authenticity of host '192.168.164.132 (192.168.164.132)' can't be established.
ED25519 key fingerprint is SHA256:L8vbU16VbbZHJ7GVdGcg5FUIEkzLgkBXqvs39dTJ4XA.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.164.132' (ED25519) to the list of known hosts.
umaiza@192.168.164.132's password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

System information as of Thu Sep 17 04:56:18 PM UTC 2026

System load:  0.0          Processes:      218
Usage of /:   32.0% of 9.75GB   Users logged in: 0
Memory usage: 11%          IPv4 address for ens33: 192.168.164.132
Swap usage:   0%

 * Canonical Workshop gives developers fast, composable, reproducible, and
   secure developer environments that are perfect for agentic workflows.

```

Key fingerprint verification prompt

Established remote SSH session displaying system status and system load.

```
Enable ESM Apps to receive additional future security updates.  
See https://ubuntu.com/esm or run: sudo pro status
```

```
umaiza@ubuntu:~$ |
```

Active remote SSH shell session.